

INDUSTRIAL PRODUCT DATASHEET

XIAO 60GHz mmWave Human Fall Detection Sensor-MR60FDA2

60GHz mmWave fall and presence sensor with XIAO ESP32C6 and Home Assistant no-code integration.

SKU

114993388

UPDATED

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PRICE

\$26.90(Excl. VAT)



Overview



The MR60FDA2 60GHz mmWave sensor detects falls within a 3x3x3 meter range and static human presence up to 6 meters, using pre-set algorithms. It features a Seeed Studio XIAO ESP32C6 pre-flashed with ESPHome, enabling no-code Home Assistant integration, along with a BH1750 ambient light sensor, WS2812 RGB LED, and Grove connector for expansion.

Key Features

Note:

Early Build provides beta products (featuring 3D-printed cases) and evolving software before final release. We welcome your feedback and suggestions to shape the product's evolution. Enjoy exploring!

Feature 1: Advanced mmWave Sensing for Fall and Human Presence Detection

The MR60FDA2 60GHz mmWave Sensor Module specializes in fall detection with pre-set algorithms and includes human static presence functionality.

- Human Static Presence: Effective up to 6 meters.
- Fall Detection: Detects within a 3x3x3 meter range, with a horizontal Field of View (FoV) of 120° and a vertical FoV of 100°.

**The above parameters have been tested in the empty deployment environments.*

Feature 2: Powered by ESP32 & ESPHome & No-Code Integration with Home Assistant

The kit features a [Seeed Studio XIAO ESP32C6](#) with pre-flashed [ESPHome](#) firmware, enabling seamless plug-and-play integration with [Home Assistant](#). Configure the device in ESPHome, and no-code setup for Home Assistant in 1-2 minutes. Thanks to the integration, you can customize detection zones and access analytics to pinpoint target locations efficiently from Home Assistant.

Feature 3: Ambient Light Sensing for Responsive Automation

The mmWave sensor module includes a BH1750 light sensor, offering precise ambient light detection from 1 to 65,535 lux, with adjustable measurements up to 100,000 lux. Supported by [ESPHome](#) and [Arduino IDE](#), it allows you to collect ambient light readings as frequently as every second, enhancing your automation's responsiveness to environmental changes.

Feature 4: Customizable Interaction

This 60GHz mmWave Sensor Module is equipped with a WS2812 RGB LED, allowing for DIY customization of the indicator light.

Feature 5: Extended Sensing or Automation with Grove

We've added one additional Grove Connector to the mmWave Module for easy expansion with your own Grove sensors and actuators (e.g., PIR and Relay). [Choose from over 300 compatible Grove Modules!](#)

Feature 6: Open Source 3D Printed Enclosure

To simplify deployment, we offer a [3D printed enclosure](#) available for free download, providing a starting point to customize your casing design.

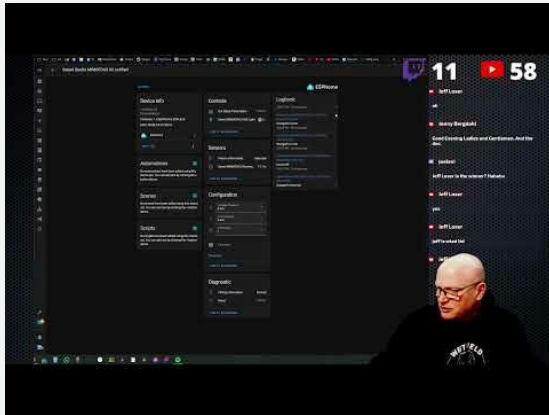
Feature 7: Open for Customization

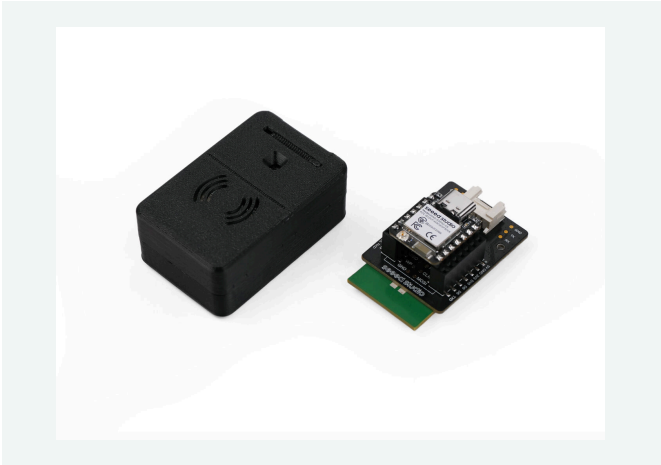
Want to tailor-make the kit to fit your unique applications?

For more information about 3D point cloud data generation and interference zone configuration when customizing mmWave modules. Seeed provides one-stop R&D customization and manufacturing services for fast development

from concept to production. Contact us at iot@seeed.cc to learn more.

Product Gallery

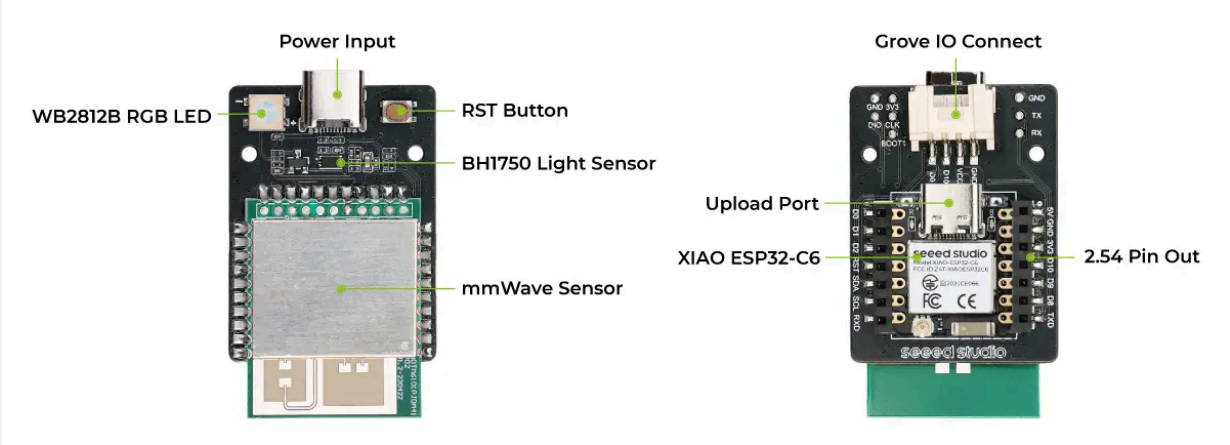




Specification

General Parameters	
mmWave Firmware	Fall Detection
Detection Range	Human Static Presence Detection: up to 6 Meters Fall Detection: • 3x3x3 meter range • Horizontal Field of View (FoV) of 120° • Vertical FoV of 100°
MCU	Seeed Studio XIAO ESP32C6
LED	WS2812 RGB LED
Button	Rest
Light Sensor	BH1750Range: 1 to 65,535 lux with adjustable measurements up to 100,000 lux
Grove Connector	1 GPIO Port • D0 • D10
Pin Header Spacing	2.54mm
Power Supply	5V/1A
Power consumption	0.5w: Standby Mode 0.8w: Activation Mode 1.4w: Work with Grove Relay status

Hardware Overview



Part List

60GHz mmWave Sensor Module with 3D Printed Case	x1
Seeed Studio XIAO ESP32C6 with Pre-Flashed ESPHome Firmware	x1

Compliance & Logistics



HSCODE	8526109090
USHSCODE	8526100040
UPC	
EUHSCODE	8471707000
COO	CHINA

FCC Requirement

Any changes or modifications not expressly approved by the party responsible for compliance could void the user's authority to operate the equipment.

This device complies with Part 15 of the FCC Rules. Operation is subject to the following two conditions:

- (1) this device may not cause harmful interference, and
- (2) this device must accept any interference received, including interference that may cause undesired operation.

15.21

This transmitter must not be co-located or operating in conjunction with any other antenna or transmitter.

Note: This equipment has been tested and found to comply with the limits for a Class B digital device, pursuant to Part 15 of the FCC Rules.

These limits are designed to provide reasonable protection against harmful interference in a residential installation.

This equipment generates, uses, and can radiate radio frequency energy.

If not installed and used in accordance with the instructions, it may cause harmful interference to radio communications.

However, there is no guarantee that interference will not occur in a particular installation.

If this equipment does cause harmful interference to radio or television reception, which can be determined by turning the equipment off and on, the user is encouraged to try to correct the interference by one or more of the following measures:

- Reorient or relocate the receiving antenna.
- Increase the separation between the equipment and receiver.
- Connect the equipment into an outlet on a circuit different from that to which the receiver is connected.
- Consult the dealer or an experienced radio/TV technician for help.

Resource Links

- <https://www.seeedstudio.com/MR60FDA2-60GHz-mmWave-Sensor-Fall-Detection-Module-p-5946.html>
- [mmWave 3D Case](#)
- [PCN-114993388.pdf](#)

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